

HIK VISION DS-7600NI-SE/P Series NVR User Manual Version 2.3.9 Starting Up and Shutting

Down Your NVR Proper startup and shutdown procedures are crucial to expanding the life of your NVR.

Startup Your NVR

1. Ensure the power supply is plugged into an electrical outlet. It is HIGHLY recommended that an Uninterruptible Power Supply (UPS) be used in conjunction with the unit.
2. Connect the NVR to a VGA/HDMI monitor. You will only see the NVR menu system when it's connected to a VGA/HDMI monitor.
3. Press the POWER switch on the back rear panel. The unit begins to start up.

Shutdown Your NVR

1. Shutdown your NVR by going to Menu > Maintenance, click in the lower left corner to pop up the Shutdown menu, as shown in Figure 2. Select the Shutdown button.
3. Click the Yes button.
4. When an attention message with words: Please power off, shown in the windows, press the POWER switch to power off.

Rebooting Your NVR

While in the Shutdown menu (Figure 7), you may also reboot your NVR by clicking Reboot.

Setting Date & Time

It is extremely important to setup the system date and time to accurately time stamp recordings and events.

Figure: System Configuration Menu

1. Enter the System Configuration menu by going to Menu > System Configuration > General.
2. Select the Time/Date tab, as shown in Figure

Locking Your NVR

Locking your NVR will return you to the Live Feed mode, which will require the correct user name and password to exit out of it and do operation. You have two methods to lock your NVR.

1. Enter the Shutdown menu by going to Menu > Maintenance > Shutdown, and click the Lock button.
2. Click on the lower left corner of Menu interface to lock your NVR. Note: You can also set auto lock time to lock the menu for a certain period of time inactivity. Enable auto locking by going to Menu > System Configuration > General > More Setting, and select auto lock time in the Auto Locktime dropdown list
3. The current system time and date as well as the time zone will be displayed. Using the directional buttons on the remote or the mouse, selects the correct date, time and time zone.
4. To enable Daylight Savings Time: Select a time zone that supports DST Check the Enable DST checkbox and click Customize button to pop up the DST Setting menu, as shown in Figure 9 You can check the Auto DST Adjusting checkbox or manually check the date of the DST period Click the Apply button to confirm the settings. Click OK to return to the upper level menu. To acquire the time and date over an NTP (Network Time Protocol) Server, check the Enable NTP checkbox. You can set the interval and enter your own NTP server.
6. Click the Apply button to save the settings and click to exit out of the menu. Clicking without clicking the Apply button will exit the menu without saving.

Checking the Status of Your NVR

The current status of your NVR can be checked at anytime by going to the System Information menu. The System Information menu can be accessed by going to Menu> System Information

1. Select the Device Info tab to enter the Device Information menu to view the device name, model, serial no., firmware version and encode version, as shown in the above Figure

Select the Camera tab to enter the Camera Information menu to view the status of each camera, as shown in the Figure pasted below

Select the Record tab to enter the Record Information menu to view the recording status and encoding parameters of each camera, as

shown in Figure given below Select the Alarm tab to enter the Alarm Information menu to view the alarm information, as shown in the following Figure Select the Network tab to enter the Network Information menu to view the network information Select the HDD tab to enter the HDD Information menu to view the HDD status, free space, capacity, etc

Configuring Live Feed Displays To customize display settings: Step 1: Set menu output resolution and mouse pointer speed by going to Menu > System Configuration > General > Display The settings available to configure in this menu include:

- Language: Resolution: Mouse Pointer Speed: Double Click Speed: The default language used is English. Select the appropriate resolution of menu output. Set the speed of mouse pointer and 4 levels are configurable. configurable.
- Set the speed of mouse double clicking, and 3 levels are Enable Password: Enable HDMI Resolution Auto: Enable or disable the password of live view screensaver. Enable or disable the auto adjust of HDMI resolution. Step 2: Set device name, device no. and auto lock time by going to Menu > System Configuration > General > More Settings, as shown in Figure The settings available to configure in this menu include:
- Device Name: Device No.: Auto Lock time: Menu Output Mode: Edit the name of the device. Edit the number of the device. Set the auto lock time of menu. The menu output mode is HDMI/VGA adaptable. Step 3: Set the live view interface parameters and event output by going to Menu > System Configuration > Live View > General, as shown in Figure The settings available to configure for each video output include:
- Video Output Interface: Live View Mode: Dwell Time: The video output interface is VGA/HDMI adaptable. Designates the display mode for Live Preview. Set the time to dwell between switching of channels when Start Auto switch is selected in Live Preview.
- Enable Audio Output: Event Output: Full Screen Monitoring Dwell Time: Enable or disable audio output for the selected video output. Designates the output to show event video. It is VGA/HDMI adaptable. when events occur.

Configuring Cameras Set the time to dwell between switching of channels

Adding IP Cameras The DS-7600NI-SE/P series NVR provides PoE interfaces which can connect to IP cameras directly. The PoE interface supports Plug-and-Play function. You can also connect to the online IP camera manually. Before configuring IP cameras, please ensure the network settings for your NVR is properly setup

To manage IP cameras: Plug-and-Play

1. Enter the Network menu by going to Menu > System Configuration > Network and check the Internal NIC IPv4 is configured properly.
2. Connect the cameras via PoE interface.
3. Enter the Cameras menu by going to Menu > Cameras Setup > Cameras
4. The cameras connect via PoE interface are listed with internal NIC IP address. Manual

1. Enter the Cameras menu as shown in Figure. Click the Search button to automatically detect connected IP cameras on the network.
3. Left-click the detected cameras. The basic information of the selected camera will display on the left side of the interface.
4. Click near the IP address to modify the camera's IP address, subnet mask and port. Input the password and click Apply to confirm the settings.
5. You can also click Protocol to pop up a Protocol Management window to edit the protocol parameters Click in the camera list shown below.

There will pop up an Edit IP Camera menu 2. 3. Select the Adding Method to Manual. Input IP camera address and other necessary information of the camera you want to connect.

4. Input the password and click the Apply button to save the settings. Click OK to exit this menu. Click Cancel without clicking Apply or OK will back without saving. 5. You can click on the camera list anytime you want to configure the added camera. 6. If the camera is manually added, you can select the camera and click the Advance Set button to pop up an Advance Settings window 7. Select the Network tab to modify the IP Camera Address and Configuration Port. Click Apply to save the settings. 8. Select Password tab to change the password. Click Apply to save the settings. 9. 10. 11. 12. Select Maintenance tab to reboot or default the selected camera. Click OK to back to the previous menu. Select Upgrade tab to upgrade IPC using firmware from external storage device. If the camera status turns Disconnected to Connected, you can click to get the live view of the camera.

Setting Camera Order

Setting the camera order allows you to logically position cameras for more efficient monitoring of your own individual location To set the camera order: 1. Enter the View menu, shown in Figure 26 by going to Menu > System Configuration > Live View > View. 2. Select the Screen Configuration you would like to use in Live Feed by clicking. The preview of the screen configuration on the right will change depending on the option selected. 3. 6. Click to select a screen in the right region and double-click to select a channel in the left region. Thus the selected channel will be displayed in the corresponding screen. 4. Click 'play' to start live view of all channels and click 'cross' to stop live view all channels. 5. Click the Apply button to save the settings. Click to exit out of the menu.

Configuring Settings for Recording

There are multiple ways to setup your NVR for recording. They include setting up a recording schedule, triggering a recording by motion detection and/or a sensor alarm.

Configuring Recording Settings

Before setting your NVR up for recording, certain settings should be configured first. The first set of settings to configure in this menu is the recording parameters. To configure the recording parameters as following steps Enter the Parameters Settings menu by going to Menu > Recording Configuration > Parameters as shown in Figure 2. Select main stream tab to configure the parameters of main stream. Select the camera to configure in the Camera dropdown list Select the Recording Mode to configure. Both Continuous and Event can be configured Select the stream type in the Stream Type dropdown list. If the camera supports audio input, the options will include Video & Audio and Video Select the camera resolution in the Resolution dropdown list. The options for the camera resolution are various according to the IP camera Select the bit rate type in the Bitrate Type dropdown list. The options for the camera bit rate type include Variable and Constant Select the recording Frame Rate to use for the designated camera. A rate of 30 (Full Frame)all the way down to 1/16 of a frame can be selected Set the video quality using the Video Quality slider. Increasing the video quality will also increase the max. bit rate recommended. The max. bit rate can be adjusted according to this recommend Select the max. bit rate mode in the Max. Bitrate Mode dropdown menu. The

options for the camera max. bit rate mode include General and Customize (32-16384kbps)

Check the checkbox of record audio to enable or disable audio while recording. If the selected camera does not support audio, the checkbox will not be editable Click the Apply button to save the settings Repeat the steps to set other cameras

Select Sub stream tab, as shown in Figure, to configure the parameters of sub stream. The steps are the same as main stream configuration

Select record tab, as shown in Figure to configure some parameters of recording

Select the Pre-record time. The pre-record time is the time in seconds to record before a recording is triggered. Setting the pre-record time to Max. will allow the NVR to use up to the maximum available buffer space for recording

Select the Post-record time. The post-record time is the time in seconds to also record after a recording has ended

Enter the Erase Video After. The Erase Video After time denotes the amount of days that files will be deleted after its initial recording. Setting the time to 0 will allow the NVR to only delete and overwrite files when the HDD is full

If check the Redundant Record checkbox, the record will be saved in the redundant HDD

Playing Back Tags

1. Enter the Tag interface, shown in Figure by going to Menu> Playback > Tag .
2. Select channel on the Channel dropdown list.
3. Select tag type for Tag Keyword or All.
4. If tag type was Keyword, Keyword area was editable to allow user to set keywords for fast searching.
5. Select start time and end time.
6. Select the pre-play and post-play time.
7. Click the Search button and the tag recording found will be listed in Result interface.
8. Click or double click the tag you want to play and it will be display in the Playback interface as shown in Figure 40.
9. Drag the timeline to the place you want to play.
10. Click to enable or disable the audio.
11. Click to play back the pictures in Full Screen mode.
12. Click to edit the tag name.
13. Click to delete the tag.

Backing up Video Clips

After video clips have been selected in the Playback Interface, you may back them up to an external USB storage device or DVD writer by going to the Clips Export menu.

Tobackup video clips:

1. Enter the Clips Export menu, by first going to Menu > Playback. In the Playback menu, click the Clips Export button. Figure: Clips Export Interface
2. If video clips were successfully saved to the HDD using the Playback Interface, they will be listed in the Clips Export interface. The camera number as well as the time range and clip size would be listed.
3. Select the video clips you would like to backup by checking the checkbox next to the desired clips. You may also click to play and review the clip. Video clips can be deleted by clicking or by clicking the Delete button to delete the selected clips. Check the Channel No. box will choose all clips.
4. 5. You can also check the checkbox of Merge files to export.

Connect at least one USB storage device to the NVR. If the device is compatible with the NVR, it will automatically be detected. Select the backup device from the Device Name dropdown menu

6. If the available space on the storage device is adequate, select the Export button to begin backup of the selected clips. After clips have been backed up, you may click the Cancel button to return to the Playback Interface.

Configuring Network Settings

Network settings must be configured before you're able to use your NVR over the network. Configuring General

Settings To configure network general settings: 1. Enter the Network Configuration menu, shown in Figure by going to Menu > System Configuration > Network 2. Select the General tab. The current network settings are displayed on the right side of the menu, shown in Figure 3. NIC Type: Set the NIC type of the device. General it is default as 10M/100M/1000M self-adaptive. 4. If you have a DHCP server running and would like your NVR to automatically obtain an IP address and other network settings from that server, check the Enable DHCP checkbox. 5. If you would like to configure your own settings, enter the settings for: IP Address: IP address you would like to use for your NVR. Subnet Mask: Subnet Mask of network. Default Gateway: IP address of your Gateway. Typically the IP address of your router. MTU: The valid value range of MTU is 500~9676. 6. 9. DNS Server: The preferred and alternate DomainName System (DNS)Server to be used with your NVR. 7. Internal IPv4 Address: You need to configure the internal NIC address, so that IP addresses are assigned to the cameras connected to the PoE or built-in switch interfaces. 8. Click the Apply button to save the settings. Click the Refresh button to refresh the updated network information after enabling DHCP. Configuring DDNS Dynamic DNS allows you to create a hostname and associate it to your IP address, making access to your NVR over the internet easier. To configure DDNS: 1. Select the DDNS tab to enter the DDNS settings interface, as shown in Figure 2. 3. Check the Enable DDNS checkbox. Select a DDNS type from the DDNS Type selection dropdown list. Five different DDNS types are selectable: IP Server, Dyn DNS, Peanut Hull, NO-IP, HiDDNS. IP Server: Dyn DNS: 1) Enter Server Address for IP Server Enter Server Address for Dyn DNS. 2) Enter the domain obtained from the Dyn DNS website in the Domain Name textbox. 3) Enter the User Name and Password registered in the Dyn DNS website. PeanutHull: NO-IP: Enter the User Name and Password obtained from the PeanutHull website 1) Enter Server Address for NO-IP. 2) Enter the domain obtained from the Dyn DNS website (www.no-ip.com) in the Domain Name textbox. 3) Enter the User Name and Password registered in the NO-IP website. HiDDNS: The Server Address defaults to www.hik-online.com Enter the Device Domain Name. The domain name can only contain the lower-case letter, numeric and '-', and it must start with the lower case letter and cannot end with '-'. If you have more than one device, you can register an account on www.hik-online.com to do some management. Changing Password You can change the default password by going to Menu > System Configuration > User To change the password: 1. Click to pop up the Edit User menu 2. Enter the old password in the Old Password textbox. 3. Check the Change Password checkbox. 4. Enter the new password in the Password textbox. 5. You can enter the User's MAC Address of the remote PC. If it is configured and enabled, it only allows the remote user with this MAC address to access the device. 6. You can also enter the User's IP Address of the remote PC. If it is configured and enabled, it only allows the remote user with this IP address to access the device. 7. Click the OK button to save the setting and back to the up level Adding a New Remote/Local User You may add up to 31 new users to your NVR. To add new users: 1. On the User interface, click

Add to enter Add User interface, as shown in Figure 54 2. Enter the information for new user, including User Name, Password, Level and User's MAC and IP Address. Level: Set the user level to Guest or Operator. Different user levels have different default operating permission. Guest: Operator: Guest has the default permission of local log search and remote log search Operator has the default permission of local log search, remote log search, two- way audio and camera configuration. 3. Click the OK button to confirm adding the user. Once a new user has been created, the fields behind Users will become editable. Changing the permission of User The Administrator can change the permission of both guest and operator. To change the permission:

1. Enter the User menu, by going to Menu > System Configuration > User. 2. Click to enter the Permission interface Set the operation permission of Local Configuration, Remote Configuration and Camera Configuration for the user. Click the Apply button to save the parameters.

Configuring PTZ Cameras Configuring Basic PTZ Settings Settings for a PTZ camera must be configured before it can be used. Before proceeding, verify that the PTZ camera is connected properly. To configure PTZ settings:

1. Enter the PTZ menu, by going to Main Menu > Cameras Setup > PTZ 2. Select the General tab. 3. Select the PTZ camera to configure in the Camera dropdown list. 4. Configure PTZ settings, including those of Baud Rate, Data Bit, Stop Bit, Parity, Flow Ctrl, PTZ Protocol and Address according to the parameters of the PTZ camera(s). 5. Click the Apply button to save the settings.

Configuring Email Settings If you would like to have the NVR send out emails when certain events are detected or exceptions have been triggered, you must first setup the email settings. To set up email settings:

1. Enter the Email Configuration menu by going to Menu > System Configuration > Network. 2. Select the Email tab 3. In this tab, enter all pertinent email information, including: Enable Server Authentication: User Name: Password: SMTP Server: SMTP Port: Enable SSL: Sender Name: Sender's Address: Enable if email server requires authentication. Enabling Server Authenticating will enable the User Name and Password fields User name to use for server authentication Password to use for server authentication Address for SMTP server Port for SMTP server Enable Secure Sockets Layer (SSL) for out-going email The sender name to use when an email is sent out from the NVR The sender's address to use when an email is sent out from the NVR Enabling will attach a small picture segment (Interval can be set below the Enable Attached Picture checkbox) to the out-going email. Receiver: Click to edit Receiver Settings. Input the receiver's name and address. Click Apply to save the settings. The email address will be added to the Receiver list. You can also delete the receiver by clicking . 4. You may now test the email settings by clicking the Test button. If the settings have been properly configured, the recipient will successfully receive a test email. 5. Select the Apply button to save the email settings and select to return to the previous menu. Selecting without clicking Save will quit out of the menu without saving settings.

Configuring Privacy Mask Privacy Mask can be setup to mask off sensitive or private areas in the field of view of a camera. To configure privacy zones:

1. Enter the Privacy Mask menu, by going to Menu > Cameras Setup > Privacy Mask. Figure: Privacy

Mask Settings Menu 2. Select the camera to setup privacy mask in using the camera dropdown list on the upper left of the menu. 3. Click Enable Privacy Mask checkbox to enable privacy mask. 4. Up to four privacy masks can be used per camera and are shown using four different colors, yellow, green, blue and grey. Using the mouse, click and drag out rectangular boxes defining the desired zones. 5. You may clear a privacy zone at any time by clicking on the corresponding Clear Zone button or Clear All button to clear all zones. 6. Select the Apply button to save the privacy mask settings and select to return to the previous menu.

Selecting without clicking Apply will quit out of the menu without saving settings. Configuring Video Loss Video loss detection can be enabled on any of the channels on your NVR to detect the loss of video. To configure video loss detection: 1. Enter the Video Loss menu, by going to Menu > Cameras Setup > Video Loss. Figure: Video Loss Detection Menu 2. Select the camera to setup video loss detection in using the camera dropdown list. 3. Check the Enable Video Loss Detection checkbox. 4. Press the Set button to set up linkage action. Select the Arming Schedule tab to set the arming schedule. Up to 8 periods can be configured. If the same schedule can be applied in other days, click Copy Select the Linkage Action tab to set the actions to take if tamper-proof is detected. The detail instruction of the actions. 5. Click the Apply button to save the settings. Click OK to back to the previous menu. 6. Click the Apply button on the Video Loss interface to save the settings.

Managing System Restoring Default Settings To restore factory default settings to your NVR: 1. Enter the Factory Default menu by going to Menu > Maintenance > Factory Default 2. Select the OK button to restore factory defaults or select to return to the previous menu. Configuration information from your NVR can be exported to a USB storage device and imported into another NVR. This will allow you to efficiently setup the same configuration on numerous NVRs. To export NVR configuration:

1. Enter the Import/Export menu by going to Menu > Maintenance > Import/Export. 2. Connect the USB storage device to a USB port on the NVR. 3. Click the Refresh button. The contents of the USB storage device will be shown on the screen. 4. You can click the New Folder button to create a new folder. 5. Select the location where you would like the configuration to be stored on the USB storage device. 6. Click Export to export a configuration file to USB storage device. The configuration file will be named dev Cfg. bin. 7. Click to exit out of the Import/Export menu.

To import NVR configuration: 1. Enter the Import/Export menu, shown in Figure 105 by going to Menu > Maintenance > Import/Export. 2. Connect the USB storage device to a USB port on the NVR. 3. Click the Refresh button. The contents of the USB storage device will be shown on the screen. 4. Select the configuration file. The configuration file is named dev Cfg. bin. 5. Click the Import button. 6. In the pop-up window click Yes to confirm the import and the system will automatically reboot to make it effective.

Viewing System Logs Many events of your NVR are logged into the system logs. To access the system logs and search for these events: 1. Enter the System Logs menu by going to Menu > Maintenance > System Logs 2. Select the Start Time and the End Time. 3. Select the Major

Type from the dropdown list. Four major types are available: Alarm, Exception, Operation and Information. You can also select All. 4. Select the Minor Type under the Major Type to efficiently find the logs. 5. Click the Search button. The search results will be displayed in a list. Click to page up and down 6. Click under the Detail item to view the detail information 7. Click Previous/Next to view the previous/next log information. 8. Click OK to back to the previous menu. 9. If applicable, you may also view the associated video to the selected log entry by clicking . 10. Log files can also be exported onto a USB storage device. To export a log file, connect a USB storage device to the NVR, select the log files to export and click the Export button. 11. Click to exit out of menu.

Network detection You can view the network traffic to obtain real-time information and connecting status of your NVR. To view network traffic:

1. Enter the Network Detection menu by going to Menu > System Maintenance > Network Detection.
2. Select the Traffic tab and you can view the sending rate and receiving rate information on the interface. The traffic data is refreshed every 1 second.

Figure: Traffic Detection

To configure network detection

1. Enter the Network Detection menu by going to Menu > Maintenance > Network Detection.
2. Select the Network Detection tab to enter the Network Detection menu.
3. Enter the destination address in the text field of Destination Address.
4. Click the Test button to start testing network average delay and packet loss rate. The testing result will pop up on the window as shown below.
5. Click the Status to check the network status. If you don't input the DNS address, there will pop up a result window as shown below.
6. You can click the Network button to check the network information and you can configure the network settings in this interface